

James Mann

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Education

University College London	Computer Science MEng	Sep 2024 – May 2027
• First year average: 81% • First Year Representative of the AI Society		

Experience

MATS	Neel Nanda's MATS Training Phase	Sep 2025 - Oct 2025
• Worked on various mini-projects including an LLM progress bar based on model internals and a project investigating LLM's latent representations of user characteristics (age, sex, star sign). • Developed a final project investigating model tendencies to reward hack, identifying that a deception direction could control gpt-oss-20b's reward hacking. • Trained linear probes to catch reward hacking intent. Reconstructed the found directions with an SAE to surface hypotheses on causes.		
Arcadia Impact	Research Analyst / Software Engineer	May 2025 -
• Led the creation of a technical tender responding to a €1,000,000 tender for AI Evaluations Infrastructure for the EU, ultimately coming second out of fourteen applying consortiums. • Part of the maintenance team for Inspect Evals, responding to issues and applying QA to prospective new Evaluations. • Independently developed cost modeling software that simulated 10,000+ instances of the company, accounting for underlying randomness, leading to 15% better resource allocation.		
Bluedot Impact	Incubator Week + Hackathon Prize	Nov 2025
• Accepted into v1 of Bluedot's Incubator Week • Spent a week developing a plan to create tailored probes for inference-time monitoring. • Came 2nd overall, winning £1000 in Bluedot's 100+ person hackathon solo developing an MVP, creating CUDA kernels to minimise latency of monitoring, eventually optimising it down to <5%.		
Impact Research Groups	Technical AI Safety Research Sprint	Feb 2025 - April 2025
• First author of a paper produced as part of an 8 week competitive research sprint, coming third. • Identified a safety-critical latent factor in high-dimensional model activation space that could steer between refusal and compliance. • Proposed a theoretical framework for the success of jailbreaks as suppressors of harmfulness perception.		

Projects

- **Non-deterministic sampling investigation**
 - Utilised multivariate regression to identify and controlling planning capabilities in model latent space.
 - Linear probes trained on input activations to predict the non-deterministic, sampling-induced distribution, utilizing the Kullback-Leibler divergence function, achieving a $0.95 R^2$.
 - Applied in-place, input activation steering to control the high-level length feature i.e. make the model output longer responses, or make the model output shorter responses, without affecting coherence.
- **Probabilistic Image Generation**
 - Implemented a denoising diffusion probabilistic model in Pytorch, trained on CIFAR100 to generate novel images from the distribution, iteratively optimising the hyperparameters to improve the plausibility of samples.
 - Followed the paper Denoising Diffusion Probabilistic Models, implementing a UNet with attention heads